

İntegrasyon Teknikleri

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$$\text{Ör: } \int (x^5 + 4x^2)^3 \cdot (5x^4 + 8x) dx = ?$$

$$\begin{aligned} \left. \begin{array}{l} x^5 + 4x^2 = u. \\ (5x^4 + 8x) dx = du. \end{array} \right\} &\Rightarrow \int (x^5 + 4x^2)^3 \cdot (5x^4 + 8x) dx = \int u^3 \cdot du \\ &= \frac{u^4}{4} + C \\ &= \frac{(x^5 + 4x^2)^4}{4} + C \end{aligned}$$

$$\text{Ör: } \int x^2 \sin x^3 dx = ?$$

$$\begin{aligned} x^3 &= u. \\ 3x^2 dx &= du. \end{aligned}$$

$$\begin{aligned} \int x^2 \sin x^3 dx &= \int \sin u \cdot \frac{du}{3} = \frac{1}{3} \cdot \int \sin u du \\ &= \frac{1}{3} \cdot (-\cos u) + C \\ &= -\frac{1}{3} \cos u + C \\ &= -\frac{1}{3} \cos x^3 + C. \end{aligned}$$

$$\underline{\text{on}} \quad \int \cot x \cdot dx = \int \frac{\cos x}{\sin x} \cdot dx = \int \frac{du}{u} = \ln|u| + C.$$

$$= \ln|\sin x| + C.$$

$$\sin x = u$$

$$\cos x \cdot dx = du$$

$$\underline{\text{on}}: \int \tan x \cdot dx = \int \frac{\sin x}{\cos x} \cdot dx = \int \frac{-du}{u} = -\ln|u| + C.$$

$$= -\ln|\cos x| + C$$

$$\cos x = u$$

$$-\sin x \cdot dx = du$$

$$\underline{\text{on}}: \int \sec x \cdot dx = ?$$

$$\sec x + \tan x = u.$$

$$(\sec x \cdot \tan x + \sec^2 x) \cdot dx = du.$$

$$\int \sec x \cdot dx = \int \sec x \cdot \frac{(\sec x + \tan x)}{\sec x + \tan x} \cdot dx$$

$$= \int \frac{\sec^2 x + \sec x \cdot \tan x}{\sec x + \tan x} \cdot dx$$

$$= \int \frac{du}{u}$$

$$= \ln|u| + C$$

$$= \ln|\sec x + \tan x| + C.$$

$$\text{Qn } \int \operatorname{cosec} x \, dx = ?$$

$$\operatorname{cosec} x + \cot x = u$$

$$(-\operatorname{cosec} x \cdot \cot x - \operatorname{cosec}^2 x) \, dx = du$$

$$\int \operatorname{cosec} x \, dx = \int \frac{\operatorname{cosec} x \cdot (\operatorname{cosec} x + \cot x)}{(\operatorname{cosec} x + \cot x)} \, dx$$

$$= \int \frac{-du}{u}$$

$$= -\ln|u| + C$$

$$= -\ln|\operatorname{cosec} x + \cot x| + C$$

$$** \int \frac{dx}{a^2+x^2} = \frac{1}{a} \arctan \frac{x}{a}.$$

$$\text{on: } \int \frac{dx}{9+x^2} = \int \frac{dx}{9 \cdot (1+\frac{x^2}{9})} = \frac{1}{9} \int \frac{dx}{1+(\frac{x}{3})^2}$$

$$= \frac{1}{9} \int \frac{3 du}{1+u^2}.$$

$$= \frac{1}{3} \arctan u + C.$$

$$= \frac{1}{3} \arctan \frac{x}{3} + C.$$

$$= \frac{1}{3} \arctan \frac{x}{3} + C$$

$$\text{on: } \int \frac{dx}{\sqrt{9-x^2}} = \int \frac{dx}{\sqrt{9 \cdot (1-\frac{x^2}{9})}} = \int \frac{dx}{3 \sqrt{1-(\frac{x}{3})^2}}.$$

$$= \frac{1}{3} \int \frac{dx}{\sqrt{1-(\frac{x}{3})^2}}$$

$$= \frac{1}{3} \int \frac{\cancel{3} du}{\sqrt{1-u^2}}$$

$$= \int \frac{du}{\sqrt{1-u^2}}$$

$$= \arcsin u + C.$$

$$= \arcsin \frac{x}{3} + C.$$

$$\frac{x}{3} = u$$

$$\frac{dx}{3} = du$$

$$\frac{x}{3} = u,$$

$$\frac{dx}{3} = du$$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \text{Arcsin} \frac{x}{a} + C$$

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Qn: $\int \frac{\sin 2x}{\sqrt{1-\sin^4 x}} dx = ?$

$$\sin^2 x = u$$

$$\frac{2 \sin x \cos x dx}{\sin 2x \cdot dx} = du.$$

$$\int \frac{\sin 2x}{\sqrt{1-\sin^4 x}} dx = \int \frac{du}{\sqrt{1-u^2}} = \arcsin u + C.$$

$$= \arcsin(\sin^2 x) + C$$

Qn: $\int \sqrt{x} \sin^2(x^{\frac{3}{2}} - 1) dx = ?$

$$x^{\frac{3}{2}} - 1 = u.$$

$$\frac{3}{2} x^{\frac{1}{2}} dx = du$$

$$\int \sqrt{x} \sin(x^{\frac{3}{2}} - 1) dx = \int \frac{2}{3} \sin^2 u du.$$

$$= \int \frac{2}{3} \cdot \frac{(1 - \cos 2u)}{2} du.$$

$$= \frac{1}{3} \int (1 - \cos 2u) du$$

$$= \frac{1}{3} \cdot \left(u - \frac{\sin 2u}{2} \right) + C.$$

$$= \frac{1}{3} \cdot \left(x^{\frac{3}{2}} - 1 - \frac{\sin 2 \cdot (x^{\frac{3}{2}} - 1)}{2} \right) + C.$$

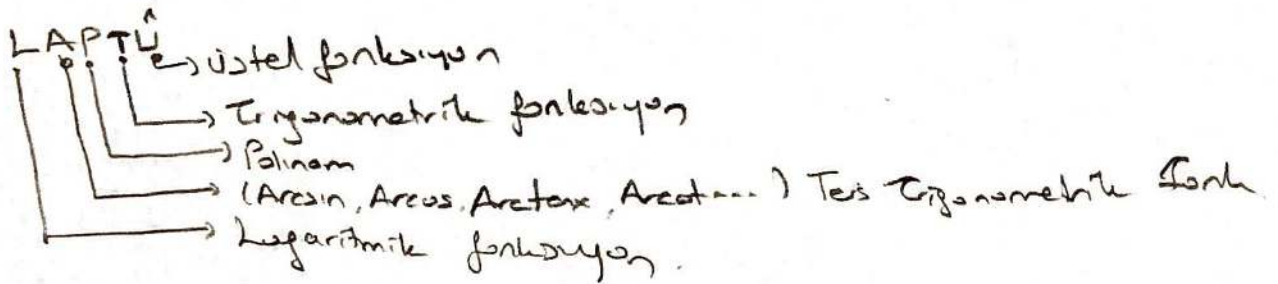
$$= \frac{1}{3} \cdot (x^{\frac{3}{2}} - 1) - \frac{\sin 2(x^{\frac{3}{2}} - 1)}{6} + C.$$

İntegrasyon Teknikleri:

Kısmi İntegrasyon

$$\int u dv = uv - \int v du$$

Belirli İntegralde: $\int_a^b u dv = uv \Big|_a^b - \int_a^b v du$



Ön: $\int x \cdot \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C$

$$\begin{aligned} & \cdot \cos x dx = dv \\ & \Downarrow \end{aligned}$$

$$\int \cos x dx = \int dv$$

$$\Downarrow$$
$$\sin x = v$$

$$\begin{aligned} & \cdot x = u \\ & \Downarrow \\ & dx = du \end{aligned}$$

$$\underline{\text{Ör:}} \int \ln x dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx = x \ln x - x + c$$

$$* \ln x = u, * dx = dv.$$

$$\Downarrow \quad \Downarrow$$

$$\frac{1}{x} dx = du, \quad \int dx = \int dv$$

$$x = v.$$

$$\underline{\text{Ör:}} \int \underbrace{x^2}_u \underbrace{e^x dx}_{dv} = ?$$

$$* x^2 = u, \quad * e^x dx = dv$$

$$2x dx = du, \quad \Downarrow \int e^x dx = \int dv$$

$$e^x = v.$$

$$\textcircled{*} \Rightarrow \int x^2 e^x dx = x^2 e^x - \int e^x \cdot 2x dx = x^2 e^x - 2 \int \underbrace{x e^x dx}_{\substack{\text{Tekrar kısmi} \\ \text{integrasyon} \\ \text{alalım.}}}$$

$$\int \underbrace{x}_u \underbrace{e^x dx}_{dv} \text{ integrali için kısmi integrasyon alalım.}$$

$$* x = u, \quad * e^x dx = dv.$$

$$dx = du, \quad \Downarrow \int e^x dx = \int dv$$

$$e^x = v$$

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x$$

$$\textcircled{*} \Rightarrow \int x^2 e^x dx = x^2 e^x - 2 [x e^x - e^x] + c = x^2 e^x - 2x e^x + 2e^x + c$$

Qn: $\int_0^4 x e^{-x} dx = ?$

* $x = u$
 $dx = du$

$\int e^{-x} dx = dv$
||
 $\int e^{-x} dx = \int dv$
 $-e^{-x} = v$

$$\int_0^4 x e^{-x} dx = -x e^{-x} \Big|_0^4 + \int_0^4 e^{-x} dx$$

$$= (-4e^{-4} - 0) - e^{-x} \Big|_0^4$$

$$= -4e^{-4} - (e^{-4} - e^0)$$

$$= -4e^{-4} - (e^{-4} - 1)$$

$$= -5e^{-4} + 1$$

$$= 1 - 5e^{-4}$$

Ön: $\int \cos^n x dx = ?$ (indirgeme form = 1 = n0 bulnu2)

$$\int \cos^n x dx = \int \underbrace{\cos^{n-1} x}_u \underbrace{\cos x dx}_{dv}$$

$$* \cos^{n-1} x = u.$$

$$- (n-1) \cos^{n-2} x \cdot \sin x dx = du.$$

$$* \cos x dx = dv$$

$\Downarrow$

$$\int \cos x dx = \int du.$$

$$\sin x = v$$

$$\int \cos^n x dx = \int \cos^{n-1} x \cos x dx = \cos^{n-1} x \sin x + (n-1) \int \sin^2 x \cdot \cos^{n-2} x dx$$

$$= \cos^{n-1} x \sin x + (n-1) \int (1 - \cos^2 x) \cos^{n-2} x dx$$

$$\int \cos^n x dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) \int \cos^n x dx$$

$$n \cdot \int \cos^n x dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx$$

$$\int \cos^n x dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x dx \quad (\text{indirgeme formül})$$

Ön: $\int \cos^3 x dx = \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \int \cos x dx$

$$= \frac{\cos^2 x \sin x}{3} + \frac{2}{3} \sin x + C.$$

Trigonometrik İntegraller

$$\int \sin^m x \cos^n x dx$$

* m tek, n çift ise.

$$u = \cos x$$
$$du = -\sin x dx$$

} denizümü ve $\sin^2 x + \cos^2 x = 1$ özdeşliği kullanarak integral çözülür.

* n tek, m çift ise.

$$u = \sin x$$
$$du = \cos x dx$$

} denizümü ve $\sin^2 x + \cos^2 x = 1$ özdeşliği kullanarak integral çözülür.

* m ve n ikisinde çift ise.

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2} \quad \left. \vphantom{\sin^2 x} \right\} \begin{array}{l} \cos 2x \text{ li d\u00fc\u0131k} \\ \text{k\u00e7terine indirgenir} \end{array}$$

* m ve n tek ise herhangi birine u der.

$$\text{Ör: } \int \sin^3 x \cos^2 x dx = \int \sin^2 x \cos^2 x \cdot \sin x dx$$

$$\cos x = u$$
$$-\sin x dx = du$$

$$= \int (1 - \cos^2 x) \cdot \cos^2 x \cdot \sin x dx$$

$$= \int (1 - u^2) \cdot u^2 \cdot (-du)$$

$$= \int (u^4 - u^2) du$$

$$= \frac{u^5}{5} - \frac{u^3}{3} + C$$

$$= \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$$

$$\hat{\text{Qn:}} \int \cos^5 x \, dx = \int \cos^4 x \cdot \cos x \, dx$$

$$\sin x = u \\ + \cos x \cdot dx = du$$

$$= \int (\cos^2 x)^2 \cdot \cos x \, dx$$

$$= \int (1 - \sin^2 x)^2 \cdot \cos x \, dx$$

$$= \int (1 - u^2)^2 \cdot du$$

$$= \int (u^4 - 2u^2 + 1) \, du$$

$$= \frac{u^5}{5} - \frac{2}{3} u^3 + u + C$$

$$= \frac{\sin^5 x}{5} - \frac{2}{3} \sin^3 x + \sin x + C$$

$$\hat{\text{Qn:}} \int \sin^2 x \cos^4 x \, dx = \int \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{1 + \cos 2x}{2} \right)^2 \, dx$$

$$= \int \left(\frac{1 - \cos 2x}{2} \right) \cdot \left(\frac{\cos^2 2x + 2\cos 2x + 1}{4} \right) \, dx$$

$$= \frac{1}{8} \int (\cos^2 2x + 2\cos 2x + 1 - \cos^3 2x - 2\cos^2 2x - \cos 2x) \, dx$$

$$= \frac{1}{8} \int (1 + \cos 2x - \cos^2 2x - \cos^3 2x) \, dx$$

$$= \frac{1}{8} \left[x + \frac{\sin 2x}{2} - \int \cos^2 2x \, dx + \int \cos^3 2x \, dx \right]$$

$$= \frac{1}{8} \left[x + \frac{\sin 2x}{2} - \frac{x}{2} - \frac{\sin 4x}{8} - \frac{\sin 2x}{2} + \frac{\sin^3 2x}{6} \right] + C = \frac{x}{16} - \frac{\sin 4x}{64} + \frac{\sin^3 2x}{48} + C$$

$$\int \cos^2 2x \, dx = \int \left(\frac{1 + \cos 4x}{2} \right) \, dx = \frac{x}{2} + \frac{\sin 4x}{8}$$

$$\int \cos^3 2x \, dx = \int \cos^2 2x \cdot \cos 2x \, dx = \int (1 - \sin^2 2x) \cdot \cos 2x \, dx = \int (1 - u^2) \cdot \frac{du}{2}$$

$$= \frac{1}{2} \cdot \left(u - \frac{u^3}{3} \right) + C = \frac{\sin 2x}{2} - \frac{\sin^3 2x}{6} + C$$

Kare Köşelerden Kurulumak

$$\text{Ön: } \int_0^{\pi/4} \sqrt{1+\cos 4x} \, dx = ?$$

$$\text{Hesaplama} \\ * \cos^2 \theta = \frac{1+\cos 2\theta}{2}$$

$$* 2\cos^2 \theta = 1+\cos 2\theta$$

$$4x=2\theta \text{ dersek. } \Rightarrow 2x=\theta \Rightarrow 2dx=d\theta.$$

$$\int_0^{\pi/4} \sqrt{1+\cos 4x} \, dx = \int_0^{\pi/2} \sqrt{1+\cos 2\theta} \cdot \frac{d\theta}{2}$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{1+\cos 2\theta} \cdot d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{2\cos^2 \theta} \, d\theta$$

$$= \frac{1}{2} \sqrt{2} \cdot \int_0^{\pi/2} \cos \theta \cdot d\theta$$

$$= \frac{\sqrt{2}}{2} \left(\sin \theta \Big|_0^{\pi/2} \right)$$

$$= \frac{\sqrt{2}}{2} \cdot (\sin \frac{\pi}{2} - \sin 0)$$

$$= \frac{\sqrt{2}}{2}.$$

$\tan x$ ve $\sec x$ Kuvvetlerinin İntegralleri

$$* 1 + \tan^2 x = \sec^2 x$$

$$\text{Ön: } \int \tan^2 x \cdot dx = ?$$

$$\text{I. yol} \int \tan^2 x \cdot dx = \int (\tan^2 x + 1 - 1) dx = \int (\tan^2 x + 1) dx - \int dx \\ = \tan x - x + C$$

$$\text{II. yol} \int \tan^2 x \cdot dx = \int (\sec^2 x - 1) dx = \int \sec^2 x \cdot dx - \int dx \\ = \tan x - x + C$$

$$\text{Ön: } \int \tan^4 x \cdot dx = \int \tan^2 x \cdot \tan^2 x \cdot dx = \int \tan^2 x (\sec^2 x - 1) dx \\ = \underbrace{\int \tan^2 x \sec^2 x \cdot dx}_{I_1} - \underbrace{\int \tan^2 x \cdot dx}_{I_2} \\ = \frac{\tan^3 x}{3} - \tan x + x + C$$

$$I_1 = \int \tan^2 x \sec^2 x \cdot dx = \int u^2 du = \frac{u^3}{3} + C = \frac{\tan^3 x}{3} + C$$

$$\tan x = u \\ \sec^2 x \cdot dx = du$$

$$I_2 = \int \tan^2 x \cdot dx = \tan x - x + C$$

$$\text{Or: } \int \sec^3 x dx = \int \underbrace{\sec x}_u \cdot \underbrace{\sec^2 x dx}_{dv}$$

$$\star \sec x = u$$

$$, \sec^2 x dx = dv$$

$$\sec x \cdot \tan x dx = du$$

$$\tan x = v.$$

$$\int \sec^3 x dx = \int \sec x \sec^2 x dx = \sec x \tan x - \int \sec x \tan^2 x dx$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx$$

$$= \sec x \tan x - \int (\sec^3 x - \sec x) dx$$

$$= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$\int \sec^3 x dx = \sec x \tan x - \int \sec^3 x dx + \int \sec x dx$$

$$2 \int \sec^3 x dx = \sec x \tan x + \int \sec x dx$$

$$2 \int \sec^3 x dx = \sec x \tan x + \ln |\sec x + \tan x| + C.$$

$$\int \sec^3 x dx = \frac{\sec x \tan x}{2} + \frac{1}{2} \ln |\sec x + \tan x| + C.$$

$$* \int \sin mx \sin nx dx, \int \sin mx \cos nx dx, \int \cos mx \cos nx dx$$

$$\sin mx \cdot \sin nx = \frac{1}{2} [\cos(m-n)x - \cos(m+n)x]$$

$$\cos mx \cos nx = \frac{1}{2} [\cos(m-n)x + \cos(m+n)x]$$

$$\sin mx \cdot \cos nx = \frac{1}{2} [\sin(m-n)x + \sin(m+n)x]$$

Ex: $\int \sin 3x \cos 5x dx = ?$

$$\int \sin 3x \cos 5x dx = \int \frac{1}{2} [\sin(-2x) + \sin 8x] dx$$

$$= \frac{1}{2} \int (-\sin 2x + \sin 8x) dx$$

$$= \frac{1}{2} \left(\frac{\cos 2x}{2} - \frac{\cos 8x}{8} \right) + C.$$

$$= \frac{\cos 2x}{4} - \frac{\cos 8x}{16} + C.$$

Trigonometrik Değişken Dönüşümleri

$$x = a \tan \theta \text{ dönüşümü, } x = a \sin \theta \text{ dönüşümü, } x = a \sec \theta \text{ dönüşümü.}$$

$$\sqrt{a^2 + x^2}$$

$$\sqrt{a^2 - x^2}$$

$$\sqrt{x^2 - a^2}$$

$$\sqrt{a^2 + a^2 \tan^2 \theta}$$

$$\sqrt{a^2 - a^2 \sin^2 \theta}$$

$$\sqrt{a^2 \sec^2 \theta - a^2}$$

$$\sqrt{a^2 (1 + \tan^2 \theta)}$$

$$\sqrt{a^2 (1 - \sin^2 \theta)}$$

$$\sqrt{a^2 (\sec^2 \theta - 1)}$$

$$a |\sec \theta|$$

$$a |\cos \theta|$$

$$a |\tan \theta|$$

$$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$0 \leq \theta < \frac{\pi}{2}, \frac{x}{a} \geq 1$$

aralığında

aralığında

$$\frac{\pi}{2} < \theta \leq \pi, \frac{x}{a} \leq -1$$

$\tan \theta$ nin
tersi vardır

$\sin \theta$ nin
tersi vardır

aralıklarında

$$x = a \tan \theta$$

$$x = a \sin \theta$$

$\sec \theta$ nin
tersi vardır

$$\theta = \tan^{-1}\left(\frac{x}{a}\right)$$

$$\theta = \sin^{-1}\left(\frac{x}{a}\right)$$

$$x = a \sec \theta$$

ya da

ya da

$$\theta = \sec^{-1}\left(\frac{x}{a}\right)$$

$$\theta = \arctan\left(\frac{x}{a}\right)$$

$$\theta = \arcsin\left(\frac{x}{a}\right)$$

ya da

dir:

dir

$$\theta = \operatorname{arcsec}\left(\frac{x}{a}\right)$$

dir.

Qn: $\int \frac{dx}{\sqrt{4+x^2}} = ?$

$x = 2 \tan \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
 $dx = 2 \sec^2 \theta d\theta$

$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2 \sec^2 \theta d\theta}{\sqrt{4+4 \tan^2 \theta}}$$

$$= \int \frac{2 \sec^2 \theta d\theta}{\sqrt{4 \cdot \underbrace{(1+\tan^2 \theta)}_{\sec^2 \theta}}}$$

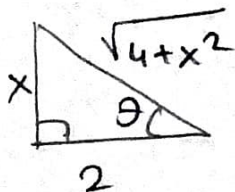
$$= \int \frac{\cancel{2} \sec^2 \theta}{\cancel{2} |\sec \theta|} d\theta$$

$$= \int \frac{\cancel{\sec^2 \theta}}{\cancel{\sec \theta}} d\theta$$

$$= \int \sec \theta d\theta$$

$$= \ln |\sec \theta + \tan \theta| + C$$

$$= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C$$



Qn: $\int \frac{x^2 dx}{\sqrt{9-x^2}} = ?$

$x = 3 \sin \theta$, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$
 $dx = 3 \cos \theta d\theta$

$$\int \frac{x^2 dx}{\sqrt{9-x^2}} = \int \frac{9 \sin^2 \theta \cdot 3 \cos \theta d\theta}{\sqrt{9-9 \sin^2 \theta}}$$

$$= \int \frac{27 \sin^2 \theta \cos \theta d\theta}{\sqrt{9 \cdot \underbrace{(1-\sin^2 \theta)}_{\cos^2 \theta}}}$$

$$= \int \frac{\cancel{27} \sin^2 \theta \cos \theta}{\cancel{3} |\cos \theta|} d\theta$$

$$= \int \frac{9 \sin^2 \theta \cancel{\cos \theta}}{\cancel{\cos \theta}} d\theta$$

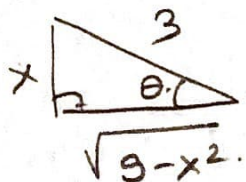
$$= \int 9 \sin^2 \theta d\theta$$

$$= \int 9 \cdot \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= \frac{9}{2} \theta - \frac{9}{4} \underbrace{\sin 2\theta}_{2 \cdot \sin \theta \cdot \cos \theta} + C$$

$$= \frac{9}{2} \arcsin\left(\frac{x}{3}\right) - \frac{\cancel{3}}{4} \cdot \cancel{x} \cdot \frac{x}{\cancel{3}} \cdot \frac{\sqrt{9-x^2}}{\cancel{3}} + C$$

$$= \frac{9}{2} \arcsin\left(\frac{x}{3}\right) - \frac{\sqrt{9-x^2}}{2} + C$$



ôñ: $\int \frac{dx}{\sqrt{25x^2-4}} = ?$

$$x = \frac{2}{5} \sec \theta.$$

$$dx = \frac{2}{5} \sec \theta \cdot \tan \theta d\theta.$$

$$\int \frac{dx}{\sqrt{25x^2-4}} = \int \frac{\frac{2}{5} \sec \theta \tan \theta d\theta}{\sqrt{4 \sec^2 \theta - 4}}$$

$$= \int \frac{\frac{2}{5} \sec \theta \tan \theta d\theta}{\sqrt{4(\sec^2 \theta - 1)}} \quad \tan^2 \theta$$

$$= \int \frac{\cancel{\frac{2}{5}} \sec \theta \cdot \cancel{\tan \theta} d\theta}{\cancel{2} \tan \theta}$$

$$= \frac{1}{5} \int \sec \theta d\theta$$

$$= \frac{1}{5} \ln |\sec \theta + \tan \theta| + C$$

$$= \frac{1}{5} \ln \left| \frac{5x}{2} + \frac{\sqrt{25x^2-4}}{2} \right| + C$$

